History of Math as a Sequence of Cognitive Revolutions

What toll did we pay to count differently? What paradoxes drove the progress?

1. Introduction

Counting is considered a trivial matter nowadays, but if we showed ancient people how we count, they would have called us witches.

- You count silently?
- You count in multitudes?
- You can count ratios?

These questions seem almost negligible to us. But what if we have actually committed a lot of cognitive steps that our ancestors would find odd and illogical?

We teach mathematics as a story of pure accumulation. The number system grows from natural numbers to integers, to fractions, to complex numbers. Each step is presented as a free upgrade - a larger box to hold everything the smaller box could hold, plus a little more.

$N \subset Z \subset Q \subset R \subset C$

But what if it is a lie?

Let's explore what it costs to invent a number.

2. Level 0. What was before counting?

We know animals can distinguish more from less. A bigger pile is more appealing to them than smaller and they get jealous. But they do not trade the same we do: A monkey will give away an apple for a grape if it wants the grape more at that moment.

This way we can imagine what our ancestors thought. Probably it was something akin to every "an apple" being "the apple", a quality rather than quantity.

Suppose a monkey sits before three apples — two green, one red — and asks itself:

— Do I want to eat? Yes. Take the sweetest food available.

So it takes the red apple. Then it sees its brother take the two green ones, and it becomes jealous. It signals:

— You took more apples than me.
— No. My apple. You ate already. I want to eat too. Prove that I took more apples than you!
— No. You clearly ate one more apple than me.
— What is one?

And then it kills its brother. (or smooches it if the brother shared one more apple with it, the exact story is shrouded in history)

That's how a natural counting starts its journey from the need to differentiate exact quantities. A story begins when humans transcended the boundaries of nature and ventured into the domain of counting the nature.

3. Level 1: Unit Counting and the Null Boundary

Now monkeys can count. And once they pick up husbandry this skill comes in handy.

A shepherd needs to know if his flock is intact. He cannot carry the sheep in his head, and he cannot trust his memory. So he picks up a pebble for every sheep that leaves the pen. He puts the pebbles in a pouch. When the flock returns, he takes a pebble out for every sheep that re-enters. If a pebble remains in the pouch, a sheep is missing.

By the standards of the Neolithic cultures, this would look like sorcery – a trick that lets you carry the flock in your pocket, forever solidifying a shepherd as a mystic figure.

This way we can imagine that basic mathematical operations available to humanity were just **2 operations:**

+1 and -1.

Tally up and tally down. Add or subtract.

The most basic operations that let you string binary flags of existence: present or absent.

In a way, these numbers do not exist until you count up to them. Moreover, there are no numbers as an abstraction - no *just 3* or *just 5*. It's always 3 apples or 5 oranges.

By learning to count, humanity paid with its position in the undifferentiated, indistinct reality, which now measures through quantifying glasses.

A primary school student cries when you ask them: "3 oranges minus 5 oranges" - and there is a very special reason behind this.

For every naturalist counter, this question is beyond stupid. Subtract 5 oranges from 3? Easy, it's Nothing (null). You had 3 oranges and then you subtracted the available oranges.

— Where did the 2 other oranges go? Nowhere, there were none to begin with, what are you talking about?!

That's when a merchant man smiles and says:
— To pay for the apples, you had to pay me 5 oranges. I will generously provide you with the apples, but you will owe me 2 oranges. Each moon, the quantity you owe me will double. See you next spring!

That's what elevates counting from natural to integers.

4. Level 2: Abstract Magnitude and Civilization

As civilizations develop trade, taxation, and construction, unit-by-unit counting becomes insufficient. The need to record quantities independently of the physical objects arises. Negative numbers are introduced as debt and are recognized as a legitimate abstraction. Nobody knows how they look, but they still use them for very exciting emergent operations.

Multiple applications of +1 and −1 create the generalized operations $+a$ and $-a$.

Then, taking multiple $+a$ and $-a$ yields multiplication ($\times a$) and division ($\div a$).

This results in exactly **4 operations**:
(+*a*; −*a*) and **(×*a*; ÷*a*)**

The numbers become an abstract magnitude that lets us count incredibly fast.

A natural counter counts apples: +1; +1; +1; +1; +1 which means "IIIII" apples.

An abstract counter counts apples: +2; +3; equals 5 apples.

But this computational power also comes at a cost. **The direct connection to reality** is forever lost.

On a ledger – the worker's salary can become -3 sheep. Every 10th sheep is taken as a tithe by the king (note that the tax is not 10%, it is "every 10th sheep").

This in theory creates the next closure problem. A paradox so trivial we sleep on it:

What is 1/10 as a number? What would become if we divide 3 by 2. We only have integers!

And in practice it boils down to:

- I came to collect the taxes. Give me every 10th sheep you have.
- I have only 32. Here are 3 sheep as is ordained by the king.
- Give me one more. 32 means there are exactly four decades of sheep ($\lceil 32/10 \rceil = 4$). This means you pay for every 10 sheep you have.
- No way. I know this is wrong, but I can't prove it! Somebody help!
- I've got an answer for you. I will teach you all about the mystic art of evading taxes. Come to my magical school to learn the secrets of the universe and explore ratios. My name is Pythagoras, by the way.

5. Level 3. Ratio and the Crime Against Numbers

To the ancient Greeks, mathematics was not a just tool for accounting. It was a philosophy of the divine. And the way they worshipped it was not with numbers, they did it through **geometry**!

Arithmetic was a messy, barbarian tool for merchants; geometry was the pure language of the cosmos that let you divide and conquer the world and nature. Lines were cut and rational ratios were measured to achieve previously unthinkable computational power.

In the context of rational numbers the quantity of operations grew in correspondence:

$(+b, -b, \times b, \div b, b^n, \sqrt{b}, |b|)$

This operation family expands to seven geometric and algebraic tools: addition, subtraction, multiplication, division, powers, roots, and absolute magnitude.

This system proved very powerful. It was the sacred math to unite all maths… Until the numbers betrayed the Greeks in a worst way imaginable.

Turns out $\sqrt{2}$ cannot be expressed through the ratio. It was a glaring stare of the Eye of Terror that showered the ancient philosophers with cold sweat and broken promises.

The bus of number universe took us for a ride and then whispered as a devil: “Kiss numbers goodbye. This is the fare to get you to the next stop”.

So, humanity ejected from the bus and entered the dark age of sciences.

$\sqrt{2}$ was declared not a *number* at all, but a *magnitude.* This forever sealed a rift between geometry and algebra, separating schools.

It was so devastating, it took a Renaissance for this curse just to be recognized. Not solved. They accepted the irrational magnitude onto the number line. And that’s when a curse even more devastating appeared: $\sqrt{-1}$.

Yes, it is no spoiler that we resolved that already, and achieve an immense computational power because of that. It even let us breach the complex realm of numbers.

But we did not patch the hole in the reality that the rational numbers created:

- Pay me 1000 guilders for this tulip bulb!
- This is outrageous!
- No, it’s ratio. Last month, it sold for 100 guilders. This month, 500. The ratio of growth is exponential. $y = 2^x$. If you buy today, next month it will be worth 2,000. You will double your investment. The curve of profits is very promising. Trust me, bro!

And then in the 21st century we continue to pay the price for a trade we did not even recognize.

6. The Fare Principle

Let’s look back at the bus ride we took. At every stop, the driver whispered the same thing:

To gain computational power, you have to sacrifice a form of certainty.

Level 1 gained distinction, but paid with the loss of indistinct reality.

- L1 → L2 gained abstract magnitude, but paid with the loss of direct reality connection.
- L2 → L3 gained ratios and roots, but paid with the loss of sign certainty.
- L3 → Complex gained complex solutions, but paid with the loss of what?

What if we sacrificed **phase certainty**?

When we accept "i", the linear number line folds into a two-dimensional plane. A number is no longer just a point with a magnitude; it has an angle, a phase. But a complex number's phase is defined modulo 2π. An angle of 0 is the exact same state as 360, or 720. The system gains immense algebraic power, but it loses the unique, linear certainty of position. The imaginary unit is the scar left by that crossing.

Now, look at the progression of operations we have built:

- **Level 1:** 2 operations (+1, -1)
- **Level 2:** 4 operations (+a, -a, ×a, ÷a)
- **Level 3:** 7 operations (+b, -b, ×b, ÷b, b^n, √b, |b|)

Each subsequent level is emergent from the operations of the previous as in a binary tree. And it seems that 1 operation is missing from level 3.

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When we solve $x^2 = 4$, the answer is x = ±2. And this is fine as an expression, but this is not an operation.

When we take the rational number into the complex realm of the curve by imposing √b — who is to say that when b = 4, √4 is actually just 2? And not 2 with some kind of indescribable numerically phase?

The root operation crosses into the complex plane, generates a hidden phase, and returns a rational number that is quietly carrying a sign-ambiguity the current system cannot express. In theory.

That is why we can actually afford to add a new operation to the list.

[[b]] = ***Obfuscation***. An operation that deliberately marks sign-uncertainty, acknowledging that a rational number may carry a hidden phase inherited from the complex plane.

7. Instead of conclusion

We now consider silent counting as easy as breathing.

But when we formalized the mathematical apparatus into the modern number line, did we forget some truths that the ancient people recognized?

The shepherd knew his pebbles were a proxy for his flock. The Greek knew that his geometry was a philosophy. The broker knew that his ratios were a gamble.

Did we actually ride the bus without paying the Fare?

Find out this and more in the full monograph.

To get you started: what if a simple matrix can tell us on operational level the emergent nature of geometry and informatics from mathematical operations following a strict binary tree pattern?

Mathematical operations:	***Geometrical objects:***	***Informatics instructions:***
2	4	6
4	8	12
8	16	24

And then this iterative pattern continues into complex math and beyond?

The answer is yes. It is renderable as a direct, emergent iteration from the ground operations of +1 and -1.

Source boundary:
This paper is a standalone synthesis of the historical and cognitive argument presented in Kirill I. Kovlekov, *Poetica Mathematica: Cogitatio ex Nihilo et Speculum Scientiae* (Version 0.25). The framework of the cognitive levels (L1–L3), the transition to the complex field, and the Fare Principle follow the monograph's internal sequence, chiefly Chapter II (§§2.2–2.9) and Chapter III (§§3.3.1). The historical mapping is presented as a structural correspondence, not a claim of direct causal determinism. The full monograph is available via Zenodo at https://doi.org/10.5281/zenodo.23698076.